

CS-1005- Discrete Structures
Semester: Spring 2026
Assignment #1
Deadline: 15th Feb 2026

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Course TAs:

Due Date: As mentioned on Google Classroom
Max Score

Note:

- 1. Please submit handwritten solutions to all of the questions given below.**
- 2. Scan your handwritten solution and upload it to Google Classroom.**
- 3. No extension in the deadline.**
- 4. Plagiarism would result in zero marks in the assignment category.**
- 5. Late submission is not acceptable.**

QUESTION 1 (55 marks)

Part 1) For each of these sets of premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion from the premises. (5 x 4 = 20 marks)

a) "If I play hockey, then I am sore the next day." "I use the whirlpool if I am sore." "I did not use the whirlpool."

b) "If I work, it is either sunny or partly sunny." "I worked last Monday or I worked last Friday." "It was not sunny on Tuesday." "It was not partly sunny on Friday."

c) "All foods that are healthy to eat do not taste good." "Tofu is healthy to eat." "You only eat what tastes good." "You do not eat tofu." "Cheeseburgers are not healthy to eat."

d) "I am either dreaming or hallucinating." "I am not dreaming." "If I am hallucinating, I see elephants running down the road."

Part 2) Show that these compound propositions are tautologies. (Truth Tables Not Allowed) (2 x 5 = 10 marks)

a) $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$

b) $((p \vee q) \wedge \neg p) \rightarrow q$

Part 3) Find the negations of these statements. (3 x 5 = 15 marks)

a) If it snows today, then I will go skiing tomorrow.

b) Every person in this class understands mathematical induction.

c) Some students in this class do not like discrete mathematics.

Part 4) Use rules of inference to show that if the premises $\forall x(P(x) \rightarrow Q(x))$, $\forall x(Q(x) \rightarrow R(x))$, and $\neg R(a)$, where a is in the domain, are true, then the conclusion $\neg P(a)$ is true. (10 marks)

QUESTION 2: The Arch Linux Catastrophe (45 Marks)

One fine Sunday, you wake up like normal, expecting to finish your pending assignments and prepare quizzes. However, when you check your email, you're bombarded with requests for help from The Linux Foundation to patch a newly discovered backdoor in the most recently released version of Arch Linux. Time is short, and stakes are high. To fix this issue once and for all, you are required to perform the following tasks:

Task 1: The Root Cause (15 marks)

The audit logs provide a set of encrypted premises regarding the breach. You cannot solve this with a simple truth table. Prove that the linker was hijacked.

You are supposed to do this question in both ways:

- 1) Rules of Inference
- 2) Tautology Method ($((p1 \wedge p2 \wedge p3 \dots) \rightarrow Q)$)

The following are the premises:

1. If the kernel was compiled with Clang (**C**) or the Linker was hijacked (**L**), then the Binary is compromised (**B**).
2. If the Binary is compromised (**B**), then the Checksum fails (**F**).
3. The root certificate is invalid ($\sim R$).
4. It is not the case that both the Kernel was compiled with Clang (**C**) and the Root Certificate is valid (**R**).
5. If the Linker was not hijacked ($\sim L$), then the Root Certificate is valid (**R**).

Task 2: The Patch (10 marks)

You find the cause responsible for the backdoor. It uses a convoluted conditional check to hide its true trigger:

$$[[(P \wedge (P \rightarrow Q)) \vee (\neg P \wedge R)] \wedge [(Q \vee R) \rightarrow P]]$$

Simplify this expression using laws of logic.

Task 3: Release Notes (10 marks)

The patch is deployed. You must write the Post-Mortem. Translate the English findings into logical notation and vice-versa.

Part A (Informal to Formal):

Translate: "There exists a developer x such that for all repositories y , if x has access to y , then y is vulnerable." (Predicates: $D(x)$, $R(y)$, $A(x,y)$, $V(y)$)

Part B (Formal to Informal):

Translate: $\forall x (Patch(x) \wedge Tested(x) \rightarrow \neg Bug(x))$

- NOTE: x is any code update or feature.

Task 4: The Suspects (10 marks)

You narrow the culprit down to three people: **Faizan**, **Umair**, and **Ahmed**.

- **Knight**: Always tells the truth.
- **Knave**: Always lies.
- **Spy**: Randomly lies or tells the truth.

The Statements:

- **Faizan**: "Umair is a Knave."
- **Umair**: "Faizan and Ahmed are the same type."
- **Ahmed**: "Umair is telling the truth."

Identify who is the Knight, who is the Knave, and who is the Spy. (You may use any method you like).

QUESTION 3: The Phantom Register (45 Marks)

(You need to show steps for each answer, assuming answers would lead to 0 marks)

A Mystery in the Machine

You're debugging a legacy microchip that's been producing inconsistent outputs. The system has been running for years, but recently started failing in unpredictable ways. Your task is to trace the logical flaw through four debugging phases.

Phase 1: The State Contradiction (5 marks)

(Requirements: Use logical equivalences and laws to simplify. Show each step and name the law used)

The system log captured this conditional check from the faulty routine:

$$\text{Check} = [(M \wedge \neg R) \vee (M \wedge R)] \wedge [\neg(\neg M \vee S) \vee R]$$

Where:

- **M**: Memory address is valid
- **R**: Register contains non-zero value
- **S**: Stack pointer is aligned

Your Task: Simplify this expression completely using logical equivalences. The final form will reveal what the system is actually checking. Write the final form with its meaning / insight as well.

Phase 2: The Execution Trace (15 marks)

(Requirements: Assign propositional variables, write premises in logical notation, apply rules of inference step-by-step, and name each rule used)

During execution, the debugger reveals these system behaviors:

1. If the watchdog timer is enabled, then either the system will reset or the timer callback was registered.
2. The timer callback was not registered.
3. The system did not reset.
4. If the DMA (Direct Memory Access) controller is active, then the watchdog timer is enabled.
5. Either the DMA controller is active or the cache is in write-through mode.

Your Task: Prove whether the cache is in write through mode. You are supposed to do this question in both ways:

- 1) Rules of Inference.
 - 2) Tautology Method $((p1 \wedge p2 \wedge p3 \dots) \rightarrow Q)$
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Phase 3: The Hardware Gate (15 marks)

(Requirements: Create the truth table and determine the circuit behavior)

You discover a suspicious custom logic gate in the circuit schematic. Its behavior is defined by:

$$\text{Output} = (A \rightarrow B) \leftrightarrow (B \wedge \neg A)$$

Where A and B are input signals.

Your Task:

- a) Construct the complete truth table for this gate.
- b) Determine if this gate could ever output True (1). If yes, under what input conditions?
- c) What type of logical expression is this (tautology/contradiction/contingency)?

Phase 4: The Memory Predicate (10 marks)

(Requirements: Translate from formal logic to clear English)

You find this assertion in the system's formal verification file:

$\forall x \ [(\text{Buffer}(x) \wedge \exists y (\text{Pointer}(y) \wedge \text{References}(y, x))) \rightarrow \text{Protected}(x)]$

Domain:

- x, y range over memory regions
- Buffer(x): "x is a buffer"
- Pointer(y): "y is a pointer"
- References(y, x): "y references x"
- Protected(x): "x has memory protection enabled"

Your Task: Translate this statement into plain English that a system programmer would understand.

QUESTION 4: The Echo of the Architect (45 Marks)

(You need to show steps for each answer, assuming answers would lead to 0 marks)

You have discovered an ancient terminal in the ruins of a secure data center. To reboot the system, you must input the 4-letter **Wake Word** by solving the logic sectors below.

Sector 1: The Council of Three (10 marks)

(Requirements: Do this part by both ways: Truth Table and by considering all possible assumptions)

The system is guarded by three nodes: **Alpha** (α), **Beta** (β), and **Gamma** (γ).

- One is a **Validator** (Always speaks Truth).
- One is a **Rogue** (Always Lies).
- One is a **Ghost** (Alternates between Truth and Lies).

The Logs:

1. α **says:** " β is the Rogue."
2. β **says:** " γ is the Validator."

3. γ **says**: " α is the Ghost."

Your Task: Identify the identity of **Node α**

- If α is the **Validator**, the code is **V**.
- If α is the **Rogue**, the code is **R**.
- If α is the **Ghost**, the code is **G**.

Sector 2: The Apex Protocol (10 marks)

An "Apex Node" (x) is a singular point of control defined by two properties:

1. It controls every node in the network.
2. No node in the network controls it.

Domain: Let N be the set of all nodes. Let $C(x, y)$ be the predicate "Node x controls Node y "

Your Task: Select the logical translation that captures this definition by **converting them all into simple English sentences**.

- **Option (O):** $\forall x [\forall y (C(x, y)) \wedge \exists z (C(z, x))]$
- **Option (I):** $\exists x [\forall y (C(x, y)) \wedge \forall z (\neg C(z, x))]$
- **Option (U):** $\exists x [\exists y (C(x, y)) \rightarrow \forall z (\neg C(z, x))]$

The Code: Use the letter of the correct option (**I**, **O**, or **U**).

Sector 3: The Casualty Chain (15 marks)

(Requirements: Use the Provided Variables. Transform the given english sentences to mathematical notations and then solve by showing each step.

YOU ARE SUPPOSED TO DO THIS QUESTION IN BOTH WAYS:

- 1) Rules of Inference
- 2) Tautology Method $((p1 \wedge p2 \wedge p3 \dots) \rightarrow Q)$

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The firewall has collapsed. Use the following "System Rules" to derive the final truth.

Variables:

- **U:** User Traffic is detected.
- **M:** Maintenance Mode is active.
- **S:** Security is online.

- **F**: Firewall is up.
- **B**: Breach is confirmed.

System Rules:

1. User Traffic is currently detected.
2. The detection of User Traffic is a sufficient condition to ensure that Maintenance Mode is NOT active.
3. It is impossible for the following three things to happen simultaneously: "Maintenance Mode is inactive, Security is online, and the Firewall is down."
4. The Firewall being up is a sufficient condition for User Traffic to be absent.
5. Security being offline is a sufficient condition for a Breach to be confirmed.

Your Task: What is the only guaranteed outcome?

- If you prove **Security** is online, the code is **S**.
- If you prove **Firewall** is up, the code is **F**.
- If you prove the **Breach** is confirmed, the code is **B**.

Sector 4: The Final Truth (5 marks)

The final lock tests the machine's core function for absolute consistency.

The Function:

$$[(p \rightarrow \neg q) \wedge (r \rightarrow q)] \rightarrow (r \rightarrow \neg p)$$

Your Task: Simplify this expression.

- If it is always True, the code is **E**.
- If it is always False, the code is **C**.
- If it depends on input, the code is **I**.

The Wake Word: [_ _ _ _] (5 marks)